

Umair Yousaf

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Education

- University of Colorado Boulder**, MS in Computer Science - CGPA 4.0 Aug 2025 – May 2027
- **Coursework:** Transformers for Robotics, Neuro-Symbolic NLP, Distributed Systems
- Lahore University of Management Sciences (LUMS)**, BS in Computer Science - CGPA 3.6 Aug 2019 – May 2023
- **Coursework:** NLP, DL, CV, ML, Data Mining, Stats & Data Analysis, Data Science, Applied Probability, HCI, SWE

Technologies

Frameworks/Tools: CUDA, HuggingFace, LangChain, LangGraph, Flask, React, Docker, MERN, Kafka, Spark, gRPC
Techniques: Vision Transformers, VQA, YOLO, OCR, RAG, Graph RAG, MCP, EDA, ETL / ETL
Languages: Python (PyTorch, TensorFlow, OpenCV), C++, JavaScript, SQL
Cloud: Azure (Azure ML, Azure Container Instances, Azure DevOps)

Experience

- Reteta**, Junior Consultant (AI Team) — Cranbury, NJ · Remote Nov 2024 — July 2025
- Performed data collection, cleaning, preparation, and **fine-tuned LLMs** to generate up to 95% accurate structured SOAP notes from patient-doctor interactions, reducing physician burnout by 75%.
 - Achieved top-tier performance in medical NLP, enhancing the product's position in the healthcare AI market.
 - Deployed and maintained a scalable ML backend on **Azure** with **Azure ML** and **Azure Container Instances**, and built CI/CD pipelines using **Azure DevOps** for automated testing, deployment, logging, and monitoring.
 - Developed **NER**-based techniques to anonymize PII, ensuring HIPAA compliance.

- Visionet-Systems**, Junior Consultant (AI Team) — Lahore · On Site July 2023 — Nov 2024
- Developed a comprehensive PDF parsing pipeline and integrated with **Agentic AI and RAG** architectures, improving document retrieval efficiency by 20%. The pipeline detected and parsed text, tables, images, captions, headers, and footers.
 - Used this pipeline to create **end-to-end Generative AI solutions** for the insurance industry, such as automated underwriting and claim management workflows to streamline business operations
 - Built a clothing recommendation system and reverse image search using transformers, Siamese networks, and contrastive learning, enhancing recommendation accuracy by 15%.
 - Created end-to-end RAG workflows using **LangChain** and **LangGraph**.
 - Trained multimodal LLMs for VQA and image-based extraction, outperforming in-house OCR.
 - Delivered live pre-sales demos to prospective clients, showcasing Generative AI solutions and building stakeholder confidence to accelerate deal progression.

- Aletheia-AI (now BrickAndMortar.AI)**, ML and SWE Intern — Lahore · On Site Jun 2022 — Aug 2022
- Deployed real-time CV systems using YOLOv5/YOLOv7 for vehicle, license plate, and person detection, achieving >90% detection accuracy for each problem.
 - Built computer vision pipelines for counting cement bags when loading onto trucks, and truck position verification on weighbridges, ensuring accurate weight measurement and significantly reducing manual intervention.
 - Managed end-to-end data workflows: collection, annotation, training, and fine-tuning of models.

Projects

Real-Time Tabletop Perception Pipeline | Robotics/Perception Project - CU Boulder

- Built a real-time RGB-D perception pipeline in ROS 2 for tabletop object detection and localization for robotic manipulation
- Integrated volumetric fusion and ESDF computation using NVBlox for collision-aware planning
- Implemented table-plane removal, robot geometry filtering, and DBSCAN-based object clustering

LEGAL-UQA: Low-Resource Urdu-English Dataset | NLP/ML Project | [arXiv link](#)

- Created the first Urdu-English legal QA dataset from Pakistan's constitution using OCR extraction, manual refinement, and GPT-4-assisted translation
- Evaluated generalist LLMs and embedding models on the dataset, achieving 99.19% human-evaluated accuracy with Claude-3.5-Sonnet; fine-tuned mt5-large-UQA-1.0 for domain adaptation
- Benchmarked retrieval performance, showing OpenAI text-embedding-3-large outperforms Mistral embeddings.

Lane Change Warning System | CV/ML Project

- Developed a dashcam video pipeline to detect vehicles (YOLOv7, ~96% mAP) and lanes (LaneNet, ~93% IoU)
- Used automatic dynamic homography to generate top-down view for lane-change detection in real time (25 FPS)